

BASE SAN MARTIN, Antarctica

WMO: 890660

Lat: 68.130S

Lon: 67.104W

Elev: 7

StdP: 101.24

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-29.8	-26.5	-35.2	0.1	-28.8	-31.4	0.2	-26.3	40.4	-16.0	30.4	-14.2	1.3	140	1.026

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.5	6.4	2.7	5.6	2.1	4.9	1.6	3.2	5.6	2.6	4.8	2.1	4.2	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1.3	4.2	3.0	0.7	4.0	2.4	0.3	3.9	2.0	15.7	5.7	14.4	5.0	13.4	4.3	6.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	22.1	17.7	DB	-31.4	8.7	4.3	1.1	-34.5	9.5	-37.0	10.1	-39.5	10.8	-42.6	11.6	(4)
(5)			WB	-31.6	5.0	4.0	0.9	-34.5	5.7	-36.8	6.2	-39.1	6.7	-42.0	7.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.4	1.8	1.0	-0.7	-3.4	-4.1	-8.6	-11.9	-11.4	-9.0	-4.9	-2.1	0.8		
	DBStd	6.74	1.67	1.94	2.71	3.45	4.21	5.81	7.48	7.51	7.24	5.08	3.45	1.95		
	HDD10.0	5258	254	253	332	402	436	557	679	664	569	463	362	286		
	HDD18.3	8300	512	486	590	652	694	807	938	923	819	722	612	544		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.7	4.0	4.4	4.8	5.5	4.8	4.9	4.4	4.7	4.9	5.0	5.1	4.5		
(15) Precipitation	PrecAvg	687	53	55	63	56	57	46	48	57	65	74	65	46		
	PrecMax	1011	98	95	109	108	113	83	86	91	92	134	107	109		
	PrecMin	463	27	30	36	16	33	20	19	15	40	39	24	11		
	PrecStd	142	22	17	14	21	18	17	16	17	14	22	25	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	6.9	6.8	5.9	6.2	4.6	4.8	3.9	5.7	6.2	6.3	7.0		
		MCWB	3.6	3.0	3.6	2.7	3.0	1.5	1.4	1.3	2.2	3.3	2.6	2.8		
	2%	DB	6.0	5.7	5.1	4.0	4.2	2.4	2.5	1.9	4.0	4.7	4.7	5.6		
		MCWB	2.2	1.9	2.1	1.0	1.5	0.4	0.1	0.3	1.2	1.9	1.4	1.8		
	5%	DB	5.2	4.7	3.9	2.2	2.8	0.2	0.1	0.3	2.5	3.5	3.5	4.7		
		MCWB	1.7	1.4	1.1	0.1	0.4	-0.7	-1.4	-0.9	0.3	1.0	0.7	1.2		
	10%	DB	4.5	3.9	2.8	1.0	1.4	-1.3	-1.9	-1.5	0.7	1.9	2.3	3.8		
		MCWB	1.3	0.9	0.5	-0.4	-0.3	-2.1	-2.8	-2.4	-0.7	0.1	0.2	0.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	3.6	3.9	2.9	3.1	2.0	1.7	1.7	2.6	3.7	3.0	3.4		
		MCDB	6.8	6.4	6.3	5.5	5.5	4.0	4.1	2.9	4.7	5.7	5.6	6.3		
	2%	WB	2.9	2.5	2.4	1.4	1.8	0.4	0.2	0.3	1.4	2.2	1.9	2.4		
		MCDB	5.1	4.6	4.6	3.3	3.7	1.9	2.2	1.6	3.6	4.3	3.9	4.8		
	5%	WB	2.2	2.0	1.5	0.5	0.7	-0.7	-1.4	-0.9	0.4	1.2	1.1	1.7		
		MCDB	4.2	3.9	3.2	1.8	2.3	0.3	-0.1	0.2	2.2	3.1	3.0	3.9		
	10%	WB	1.8	1.4	0.8	-0.3	-0.3	-2.0	-2.9	-2.4	-0.7	0.1	0.4	1.1		
		MCDB	3.7	3.2	2.3	0.9	1.2	-1.4	-1.9	-1.5	0.7	1.8	2.1	3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.5	3.9	3.9	4.3	5.3	6.9	7.5	7.4	6.2	5.1	3.8		
		MCDBR	4.7	4.7	5.4	5.2	5.6	6.2	7.8	8.3	7.2	6.8	5.6	4.9		
	5% WB	MCWBR	3.0	3.0	3.7	3.7	4.0	5.1	6.8	7.2	5.5	5.2	3.8	3.2		
		MCWBR	4.1	4.2	4.8	4.9	5.2	6.0	7.9	7.9	7.2	6.7	5.6	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.263	0.243	0.201	N/A	N/A	0.140	0.196	0.236	0.251	0.259	0.269		
		taud	2.328	2.361	2.366	2.219	N/A	N/A	1.908	2.098	2.224	2.226	2.255	2.273		
	Ebn at Noon		982	929	842	632	N/A	0	135	603	822	930	987	998		
		Edh at Noon	107	89	66	37	N/A	0	6	40	74	101	114	117		
(44) All-Sky Solar Radiation	RadAvg		7.21	4.75	2.29	0.64	0.08	0.00	0.02	0.38	1.62	4.19	6.81	8.33		
	RadStd		0.32	0.31	0.19	0.06	0.01	0.00	0.00	0.03	0.14	0.23	0.31	0.41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page